

Ella Lewis

438-927-0678 | elcslewis@gmail.com | English/French

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Verilog-A, VHDL, Assembly
Tools and Technologies: KiCAD, MatLab, SPICE, Keysight IC-CAP, TCAD
Relevant Coursework: Semiconductor Chemistry I, Electronics Circuits, Digital Computers I

PUBLICATIONS

- **Ella Lewis**, Rebecca Fang, Daiyao Xu, Johan Alant, Ryan Griffin, Lan Wei, “RF GaN HEMT Compact Modelling with High-Temperature Effects for Extreme Environment Applications,” *21st Canadian Semiconductor Science and Technology Conference (CSSTC)*, 2026.

EXPERIENCE

Product Applications Engineering Intern Aug 2025 – Present
Infineon Technologies Inc. | *PSPICE, PLECs, Altium, Matlab, TCAD* *Ottawa, ON*

- Characterized **GaN-on-Si** integrated power-stage devices using electro-thermal **SPICE** models, extracting temperature-dependent parameters including **nonlinear capacitance variation**, **$R_{DS(on)}$** degradation, and protection circuitry behavior.
- Performed electrical characterization of GaN power devices under **DPT**, **ZVS**, and continuous switching conditions, investigating dynamic **$R_{DS(on)}$** degradation, **I_{DSAT}** shifts, and thermal effects from **-50°C to 150°C**.
- Conducted electrical stress testing on GaN power devices to investigate **PXP degradation**, evaluating **performance degradation and reliability** impacts through post-stress device characterization.
- Partnered with the Failure Analysis team to investigate GaN HEMT devices using **SEM imaging**, cross-sectioning, and optical/photon microscopy, supporting root-cause analysis, competitor benchmarking, and failure mechanism assessment.

Undergraduate Research Assistant (GaN RF e-HEMT) January 2026 – May 2026
University of Waterloo, MIT Virtual Source Gate (MVSG) | *Keysight IC-CAP* *Waterloo, ON*

- Developed a temperature-aware compact model for **GaN RF e-HEMTs** under Prof. Lan Wei using the **MIT MVSG** framework, performing parameter extraction and nonlinear curve fitting on measured device data via Keysight IC-CAP.
- Assisted in **wafer-level probing station measurements**, acquiring on-wafer DC and bidirectional device data used directly for compact model parameter extraction.
- Calibrated Gate-Leakage model parameters to sustain predictive accuracy across temperature sweeps, improving MVSG reliability for **high-power, high-frequency** RF applications in defense and space environments.

Electrical Distribution Intern Jan 2025 – May 2025
EQUANS (Plan-Group) | *Python, AutoCAD, Revit, Bluebeam* *Vancouver, BC*

- Validated **70% of 600V–1.2kV MV** substation infrastructure by cross-referencing short-circuit and arc flash studies, enabling detection of **30+** transformer impedance and breaker miscoordination issues.
- Performed accurate **load calculations** for feeders and branch circuits, ensuring **NEC code-compliant** conductor sizing and correcting discrepancies in panelboard schedules for enhanced safety and system efficiency.
- Built a **Python-based QA/QC automation tool** to continuously audit LV and MV board designs against spec sheets, improving detection of non-compliant components to **95% accuracy**, enabling project milestone tracking.

PROJECTS

mmWave Sensor Detection System | *Embedded C, KiCAD* May 2026 – July 2026

- Designed full schematic and PCB for a **24 GHz mmWave [LD2410] + PIR dual-sensor** presence detection system interfaced with an ESP32-S3.
- Integrated **battery management** (TP4056 Li-ion charger) and LDO power regulation [AP2112K], designing decoupling networks and power sequencing for mixed-signal reliability.
- Routed high-frequency UART/SPI signal lines with attention to **signal integrity**, boot configuration, and RF antenna clearance for the mmWave sensor.

EDUCATION

University of Waterloo Waterloo, ON
B.Eng Honours Electrical Engineering Co-op, President's Scholarship, 3.95 GPA *September 2024 – Present*